

HEALTHCARE ANALYTICS

Patient Flow Dashboard — Project Report

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Dataset: Healthcare Patient Flow (9,216 records | Apr 2023 – Oct 2024)

Tools: Excel · MySQL · Power BI

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1. Executive Summary

This report presents the findings of an end-to-end healthcare analytics project analyzing patient flow data from a hospital system covering April 2023 to October 2024. The dataset contains 9,216 patient visit records across 8 departments (including walk-in patients), capturing demographics, wait times, admission flags, and satisfaction scores.

The project was delivered in three phases: data cleaning and documentation in Excel, exploratory and analytical querying in MySQL, and an interactive dashboard presentation in Power BI. Findings reveal critical operational bottlenecks around wait time performance, gaps in patient satisfaction data collection, and demographic patterns in admission rates.

Key Performance Indicators

Metric	Value
Total Patient Visits	9,216
Date Range	April 2023 – October 2024 (19 months)
Admission Rate	50.0% (4,612 admitted)
Average Wait Time	35.3 minutes
Delayed Wait Rate (>30 min)	80.8%
Average Satisfaction Score	4.99 / 10
Satisfaction Survey Fill Rate	27.3% (significant gap)
Walk-In / No Referral Visits	58.6% of all visits

2. Project Methodology

2.1 Phase 1 — Data Cleaning (Excel)

The raw CSV was imported into Excel and subjected to a structured data audit. All issues were logged in a dedicated Data Issues Log sheet to maintain full transparency and reproducibility. The following problems were identified and resolved:

#	Column	Issue Found	Action Taken	Rows
1	Admission Date	Mixed date formats (YYYY-MM-DD and DD/MM/YYYY)	Standardised to YYYY-MM-DD using dayfirst parsing	9,216
2	Patient Gender	Typo value "Femaleemale"	Replaced with "Female"	1
3	Department Referral	5,400 NULLs (58.6% missing)	Filled with "Walk-In / No Referral"	5,400
4	Satisfaction Score	6,699 NULLs — only 27.3% recorded	Retained as NULL; flagged for data collection improvement	6,699
5	Duplicate Columns	4 lowercase duplicates of existing columns	Removed duplicate columns	—
6	Admission Date	Exact copy of Patient Admission Date	Removed	—
7	Unnamed: 20	Residual raw date string column	Removed	—
8	Admission Status	All rows = "Not Admitted" (zero variance)	Retained; flagged as low-utility for analytics	9,216

The cleaned workbook contains six sheets: Cleaned Data (formatted table with filters), Data Issues Log, Summary Statistics, and three pivot summary sheets covering department wait times, demographics, and monthly trends.

2.2 Phase 2 — SQL Analysis (MySQL)

A MySQL database was created and the cleaned dataset was imported. 25+ queries were written across six thematic sections, demonstrating proficiency in aggregate functions, GROUP BY, CTEs, window functions (LAG, RANK, NTILE, running totals), and reusable views.

Three views were created specifically for Power BI connectivity: vw_monthly_kpi, vw_dept_performance, and vw_demographic_summary.

2.3 Phase 3 — Power BI Dashboard

The Power BI dashboard was built with four interactive pages connected via cross-page slicers. The dashboard is designed for a mock executive audience — hospital administrators and department heads — and emphasises actionable insights over raw data display.

3. Key Findings

3.1 Wait Time Performance — Critical Issue

The most significant finding in this dataset is the pervasive problem with wait time performance. 80.8% of all 9,216 patient visits are classified as "Delayed" (wait time exceeding 30 minutes), meaning only approximately 1 in 5 patients receives care within the target threshold.

Metric	Value
Fast Visits (≤ 30 min)	1,771 (19.2%)
Delayed Visits (> 30 min)	7,445 (80.8%)
Average Wait Time	35.3 min
Minimum Wait Time	10 min
Maximum Wait Time	60 min

By department, Neurology (avg 36.8 min) and Physiotherapy (avg 36.6 min) have the longest average wait times, while Renal (avg 34.7 min) performs best among referred specialties. Walk-in patients, who make up 58.6% of all visits, experience similar wait times to referred patients, suggesting the bottleneck is systemic rather than specific to any intake pathway.

3.2 Patient Satisfaction — Data Quality Gap

The average patient satisfaction score is 4.99 out of 10 — effectively at the midpoint, suggesting a large portion of patients have neutral-to-negative experiences. However, this finding must be interpreted cautiously: only 2,517 of 9,216 visits (27.3%) have a satisfaction score recorded.

This 72.7% missing rate means satisfaction data cannot reliably represent the full patient population. Department-level satisfaction averages are based on small samples and should not be used for benchmarking without first addressing the collection gap.

Among the data that exists, Senior patients (age 60+) report the lowest average satisfaction at 4.77 out of 10, compared to 5.09 for Adult patients. This is a meaningful signal that the senior patient experience may require targeted improvement.

3.3 Admission Patterns

The overall admission rate is 50.0%, split nearly evenly between admitted (4,612) and not admitted (4,604) patients. Admission rates are remarkably consistent across all racial/ethnic groups, ranging from 49.1% to 51.6%, providing no evidence of demographic disparity in admission decisions.

It is important to note that the Admission Status column contains only "Not Admitted" for all 9,216 records despite the Admission Flag column showing a 50% admission rate. This discrepancy suggests the Admission Status column was either not populated correctly or reflects a different status concept — an additional data quality concern to raise with the source system team.

3.4 Volume & Trends

Patient visit volume is highly stable across the 19-month observation period, averaging approximately 485 visits per month with no strong seasonal pattern. Monthly volume ranges from a low of 431 (February 2024) to a high of 530 (August 2024) — a relatively narrow band.

Walk-in patients (those arriving without a departmental referral) account for 5,400 of 9,216 visits (58.6%), making this the single largest patient flow category. Among referred patients, General Practice sends the most referrals (1,840), followed by Orthopedics (995).

3.5 Demographic Profile

The patient population is diverse. White patients represent the largest group (27.9%), followed by African American (21.2%), Two or More Races (16.9%), Asian (11.5%), and Declined to Identify (11.2%). Wait times are consistent across racial groups (all between 34.8–36.0 min), and admission rates show no statistically meaningful disparity.

By age group, the patient population is majority Adult (53.6%), with Seniors accounting for 25.0% and Children for 21.4%. Children experience the longest average wait time (35.5 min) though the difference across groups is minimal.

4. Recommendations

4.1 Operational Recommendations

R1: Implement a Fast-Track Pathway for Walk-In Patients [HIGH PRIORITY]

With 80.8% of patients waiting longer than 30 minutes and 58.6% arriving as walk-ins, a triage fast-track system for low-acuity walk-in patients is the single highest-impact intervention available. Target: reduce the delayed rate from 80.8% to below 60% within 6 months of implementation.

- Introduce a nurse-led triage at the point of walk-in registration
- Create a designated fast-track lane for patients with minor complaints
- Set a KPI target of ≤ 30 min average wait for fast-tracked patients

R2: Resource Audit for Neurology and Physiotherapy

These two departments consistently show the longest average wait times. A staffing and scheduling audit should be conducted to determine if capacity matches demand.

- Analyse appointment scheduling data to identify bottlenecks
- Consider staggered shift starts to cover peak admission hours
- Evaluate feasibility of adding one additional slot per half-day session

4.2 Data Quality Recommendations

R3: Mandate Satisfaction Score Collection [HIGH PRIORITY]

A 72.7% missing rate for satisfaction scores makes this metric essentially unusable for decision-making. This must be treated as a system-level data collection failure, not a data analysis limitation.

- Integrate satisfaction surveys directly into the patient discharge workflow
- Set a minimum target of 70% completion rate within 3 months
- Consider a post-visit SMS/email survey as a supplementary channel
- Report satisfaction fill rate as a separate KPI on the dashboard

R4: Resolve Admission Status Column Discrepancy

The Admission Status column shows "Not Admitted" for 100% of records, conflicting with the Admission Flag column which shows a 50% admission rate. This needs to be escalated to the source system team to clarify the column definitions and correct data entry processes.

R5: Implement a Walk-In Pre-Registration System

A web or kiosk-based pre-registration system would allow walk-in patients to register their visit and basic complaint before physically arriving, reducing front-desk processing time and enabling better capacity planning.

4.3 Dashboard & Analytics Recommendations

R6: Establish Monthly KPI Review Cadence

The Power BI dashboard should be reviewed by department heads monthly. A standardized KPI scorecard covering wait time, satisfaction score, admission rate, and volume should be distributed before each review meeting.

- Monthly review meeting with department heads (first Tuesday of each month)
- Automated Power BI email report subscription to hospital administrator
- Quarterly trend analysis to identify any emerging seasonal patterns

5. Power BI Dashboard Guide

The dashboard is structured as a four-page interactive report designed for a mock presentation to hospital leadership. All pages share a common slicer panel for date range, department, gender, age group, and race.

Page 1 — Executive Summary

Target audience: Hospital CEO, CFO, COO. This page provides a single-screen snapshot of the four headline KPIs with trend indicators, a monthly volume line chart, an admission flag donut chart, and a bar chart of visits by department. The goal is to answer "How are we doing overall?" in under 60 seconds.

Page 2 — Wait Time Deep Dive

Target audience: Operations Manager, Department Heads. This page centers on a gauge visual showing average wait time against the 30-minute target (currently at 35.3 min — above target). Supporting visuals include a department bar chart with % delayed labels, a wait time heatmap by day of week, and a clustered bar chart breaking down Fast vs Delayed by department.

Page 3 — Patient Satisfaction

Target audience: Patient Experience Team, Quality Manager. This page leads with a scatter chart plotting individual wait time vs satisfaction score to test the correlation hypothesis. A satisfaction-by-department bar chart, a distribution histogram of scores, and a matrix of satisfaction by race and age group support deeper analysis. Prominently features the 27.3% fill rate as a data quality alert.

Page 4 — Demographic Breakdown

Target audience: Equity & Inclusion Officer, Medical Director. This page uses a race-by-admission-rate clustered bar chart, an age group breakdown, a gender split donut, and a wait time comparison across demographic segments. The page is designed to detect and communicate any disparities in care access or quality.

Slicer Configuration

All four pages include synced slicers for: Date Range (AdmissionDate), Department (DeptReferral), Gender, Age Group (AgeGroup), and Race. The slicers are configured in the Power BI Sync Slicers panel to propagate across all pages simultaneously.

6. GitHub Repository Structure

The project is published to GitHub as a portfolio project demonstrating end-to-end data analytics capability. The repository is structured to be self-contained and reproducible by any analyst with access to the tools used.

File / Folder	Description
data/healthcare_analytics_patient_flow_data_raw.csv	Raw dataset as received
excel/Healthcare_Patient_Flow_Cleaned.xlsx	6-sheet cleaned workbook with audit log
insights/dash.jpg	Screenshot of the dashboard
powerbi/dash	Power BI dashboard file
sql/healthcare_mysql_queries.sql	Full SQL library (25+ queries, 3 views)
report/Healthcare_Analytics_Project_Report.pdf	This report
README.md	Project overview, methodology, insights, setup guide

The README.md is written to function as a standalone showcase document for recruiters and hiring managers, covering the business questions, data cleaning steps, SQL sections, dashboard layout, key insights, and reproduction instructions without requiring the reader to open any files.

7. Conclusion

This project demonstrates a complete, professional-grade data analytics workflow applied to a real-world healthcare operational dataset. Starting from a raw CSV with multiple data quality issues, the project delivers a cleaned, well-documented dataset, a comprehensive SQL analysis library, and an interactive Power BI dashboard aligned to business objectives.

The most critical finding — that 80.8% of patients experience wait times exceeding the 30-minute target — represents an actionable operational problem that hospital leadership can address with the specific recommendations provided in Section 4. Equally important is the 72.7% missing rate for patient satisfaction scores, which must be resolved before satisfaction metrics can be relied upon for quality management.

The consistency of admission rates across demographic groups is a positive finding, suggesting equitable access to inpatient care. However, the disparity in satisfaction scores for Senior patients (4.77 vs 5.09 for Adults) warrants targeted investigation and improvement efforts.

With the infrastructure established in this project — cleaned data, SQL views, and a live Power BI dashboard — the hospital is positioned to move from reactive data review to proactive, KPI-driven management of patient flow operations.

End of Report